

Mathematical Foundations of Machine Learning Midterm 1

Time: 90 minutes

Total: 100 points

Question 1 (20 points)

Let

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ 1 & 1 & 0 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}.$$

1. (8 points) Solve the system $\mathbf{Ax} = \mathbf{b}$ and describe all solutions $\mathbf{x} \in \mathbb{R}^3$.
2. (6 points) Find a nonzero vector $\mathbf{y} \in \mathbb{R}^3$ such that $\mathbf{y}^\top \mathbf{A} = \mathbf{0}^\top$. Use your $\mathbf{y}$ to state a necessary condition on $\mathbf{b}$ for $\mathbf{Ax} = \mathbf{b}$ to be solvable, and verify that the given $\mathbf{b}$ satisfies this condition.
3. (6 points) Compute $\text{rank}(\mathbf{A})$ and $\text{rank}([\mathbf{A} \ \mathbf{b}])$.

Question 2 (15 points)

Let

$$\mathbf{u}_1 = [1 \ 2 \ -1]^\top, \quad \mathbf{u}_2 = [2 \ 1 \ 0]^\top, \quad \mathbf{x} = [1 \ 0 \ 2]^\top \in \mathbb{R}^3,$$

and let $U = \text{span}\{\mathbf{u}_1, \mathbf{u}_2\} \subseteq \mathbb{R}^3$.

1. (8 points) Find a basis for the orthogonal complement $U^\perp$ (with respect to the standard dot product). Also state $\dim(U)$ and $\dim(U^\perp)$.
2. (4 points) Find the unit vector $\mathbf{u}' \in U^\perp$ (i.e., $\|\mathbf{u}'\|_2 = 1$) that is closest to $\mathbf{x}$ in Euclidean distance, i.e.,

$$\mathbf{u}' = \arg \min_{\mathbf{z}} \{\|\mathbf{x} - \mathbf{z}\|_2^2 : \mathbf{z} \in U^\perp, \|\mathbf{z}\|_2 = 1\}.$$

3. (3 points) Compute $\cos(\omega)$, where ω is the angle between $\mathbf{x}$ and the vector $\mathbf{u}'$ found in part (2).

Question 3 (15 points)

1. (6 points) For each set below, decide whether it is a subspace of $\mathbb{R}^3$.

$$U_1 = \{\mathbf{x} \in \mathbb{R}^3 : x_1 - 2x_2 + x_3 = 0\}, \quad U_2 = \{\mathbf{x} \in \mathbb{R}^3 : x_1 - 2x_2 + x_3 = 1\}, \quad U_3 = \{\mathbf{x} \in \mathbb{R}^3 : x_1x_2 = 0\}.$$

2. (6 points) Let

$$\mathbf{v}_1 = [1 \ 0 \ 1 \ 2]^\top, \quad \mathbf{v}_2 = [2 \ 1 \ 3 \ 5]^\top, \quad \mathbf{v}_3 = [1 \ 1 \ 2 \ 3]^\top \in \mathbb{R}^4.$$

Find a basis of $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$.

3. (3 points) Let

$$\mathbf{w} = [3 \ 1 \ 4 \ 7]^\top.$$

Express $\mathbf{w}$ as a linear combination of the basis vectors you found above and state the dimension of the span.

Question 4 (15 points)

Let $\Phi : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear map. Let $B = (\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3)$ be the standard basis of $\mathbb{R}^3$ and $C = (\mathbf{e}_1, \mathbf{e}_2)$ the standard basis of $\mathbb{R}^2$. Suppose

$$\Phi(\mathbf{e}_1) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \Phi(\mathbf{e}_2) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \quad \Phi(\mathbf{e}_3) = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

1. (6 points) Find the transformation matrix $\mathbf{A}_\Phi$ of Φ with respect to B and C .
2. (3 points) Let $\mathbf{x} = [1 \ 2 \ -1]^\top$. Compute $\mathbf{y} = \Phi(\mathbf{x})$.
3. (6 points) Find a basis for $\text{Ker}(\Phi)$ and a basis for $\text{Im}(\Phi)$, and verify the rank-nullity theorem.

Question 5 (10 points)

Let $V = W = \mathbb{R}^2$. Let $B = C = (e_1, e_2)$ be the standard basis. Define

$$\tilde{B} = (\tilde{b}_1, \tilde{b}_2), \quad \tilde{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \tilde{b}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \tilde{C} = (\tilde{c}_1, \tilde{c}_2), \quad \tilde{c}_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad \tilde{c}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Suppose $\Phi : V \rightarrow W$ has transformation matrix (with respect to B and C)

$$A_\Phi = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

- (4 points) Find matrices S and T such that for every $x \in V$ and $y \in W$,

$$\alpha = S\tilde{\alpha}, \quad \beta = T\tilde{\beta},$$

where α (respectfully $\tilde{\alpha}$) denotes the coordinate vector of x in basis B (respectfully $\tilde{B}$), and β (respectfully $\tilde{\beta}$) denotes the coordinate vector of y in basis C (respectfully $\tilde{C}$).

- (4 points) Compute the transformation matrix $\tilde{A}_\Phi$ of Φ with respect to $\tilde{B}$ and $\tilde{C}$.
- (2 points) Are A_Φ and $\tilde{A}_\Phi$ similar, equivalent, or neither? Briefly justify.

Question 6 (10 points)

- (3 points) Let

$$G = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \langle x, y \rangle_G := x^\top G y, \quad \|x\|_G := \sqrt{\langle x, x \rangle_G}.$$

Compute $\langle x, y \rangle_G$ and $\|x\|_G$ for

$$x = \begin{bmatrix} 1 & -1 \end{bmatrix}^\top, \quad y = \begin{bmatrix} 2 & 0 \end{bmatrix}^\top.$$

Also determine whether x is G -orthogonal to $z = \begin{bmatrix} 1 & 1 \end{bmatrix}^\top$.

- (4 points) Let $U = \text{span}\{v_1, v_2\} \subseteq \mathbb{R}^3$ where

$$v_1 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}^\top, \quad v_2 = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix}^\top.$$

Apply Gram-Schmidt (with the standard dot product) to produce an orthonormal basis of U .

- (3 points) Using your orthonormal basis, compute the orthogonal projection of

$$x = \begin{bmatrix} 2 & 1 & 3 \end{bmatrix}^\top$$

onto U , and compute the projection error.

Question 7 (15 points)

Let $G \in \mathbb{R}^{n \times n}$ be symmetric positive definite. Define

$$\langle u, v \rangle_G := u^\top G v, \quad \|u\|_G := \sqrt{u^\top G u}.$$

Let $U = \text{span}(b_1, \dots, b_m) \subseteq \mathbb{R}^n$ where $B = [b_1 \ \dots \ b_m] \in \mathbb{R}^{n \times m}$ has full column rank.

- (5 points) Let $U = \text{span}(B)$ and consider the optimization problem

$$\alpha^* = \arg \min_{\alpha \in \mathbb{R}^m} \|x - B\alpha\|_G^2.$$

Derive the normal equations

$$B^\top G B \alpha^* = B^\top G x.$$

Hint: the G -orthogonal projection minimizes the distance to x over all vectors in U

- (5 points) Using your result, derive a matrix P_G such that the G -orthogonal projection of x onto U is $\eta_U^{(G)}(x) = P_G x$, and prove that $P_G^2 = P_G$.
- (5 points) Let $r = x - P_G x$. Prove that r is G -orthogonal to U (i.e., $B^\top G r = 0$) and explain why this implies the projection is unique.